

FINAL EXPLANATION: MATHEMATICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: MATHEMATICS | GRADE 10 | SUB-STRAND 2.1: SIMILARITY AND ENLARGEMENT

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

Driving question: Why does resizing a passport photo change its size but never its shape, and what is the mathematics that makes this possible? Use the Kimathi Street passport photo story (a 35 mm $\times$ 45 mm print and a 105 mm $\times$ 135 mm print of the same image) to answer every section. Write in full sentences, show every step of your working, and include units. Use the words from our lessons: linear scale factor (LSF), area scale factor (ASF), volume scale factor (VSF), corresponding sides, corresponding angles and similar. Write your answers in the space under each prompt.

Part 1: Finding the Number that Connects the Two Prints

A student in Nairobi has two prints of the same photo made at a studio along Kimathi Street. The small print is 35 mm wide and 45 mm tall. The large print for the school noticeboard is 105 mm wide and 135 mm tall.

- (a) Calculate the ratio of the large width to the small width, and the ratio of the large height to the small height. Show your working.
- (b) What do we call this number? Write the formula you used to find it.
- (c) Is this an enlargement or a reduction? How do you know from the value of the number?
- (d) The face on the small print has an eye-to-chin distance of 20 mm. Use the number from part (a) to find the eye-to-chin distance on the large print.

Write your answer here:

Part 2: Why the Shape Is Kept, and Why a Distorted Photo Is Not Similar

The photographer accidentally makes a third print by tripling the width of the small photo but leaving the height at 45 mm. The face looks squashed and unrecognisable.

- (a) Write the width and height of the distorted print and calculate the ratio of new width to old width and the ratio of new height to old height.
- (b) State the two conditions that must both be true for two figures to be similar.
- (c) Use your answer to (a) and the conditions in (b) to explain why the distorted print is not similar to the small print, while the 105 mm × 135 mm print is.
- (d) Explain why similarity is a multiplicative relationship and not an additive one. Give one example from the photos.

Write your answer here:

Part 3: From Lengths to Area and Volume

- (a) Calculate the area of the small print and the area of the large print in mm^2 . Then find the ratio of the large area to the small area.
- (b) Using a small rectangle with sides l and w that is enlarged by LSF k , write an algebraic proof showing how the area changes. State the relationship between ASF and LSF.
- (c) The photo had been a three-dimensional object instead of a flat print, and every length was multiplied by 3. State the relationship between VSF and LSF and calculate the VSF.
- (d) A student says: 'The large print is 3 times as big, so it uses 3 times as much paper.' Explain what is wrong with this statement.

Write your answer here:

Part 4: Choosing the Correct Scale Factor in Kenyan Problems

Before you calculate, state which scale factor (LSF, ASF or VSF) the problem gives you and which one you need.

- (a) A maize plot in Kericho is represented by a plan. The real plot is similar to the plan, with LSF = 5 from plan to real plot. The plan has an area of 120 m^2 . Find the area of the real plot.
- (b) A road sign near Nakuru is similar to a model sign. The model sign has an area of 900 cm^2 and the real sign has an LSF of 4 compared with the model. Find the area of the real sign.
- (c) A small water tank on a Kericho tea farm holds 40 litres. A similar larger tank has LSF = 3 compared with the small one. Find the capacity of the larger tank.
- (d) A map of Nairobi to Nakuru uses a scale of 1 : 500 000. The distance between the towns on the map is 18 cm. Find the real distance in kilometres.

Write your answer here:

Part 5: The Final Explanation

Write a complete answer to the driving question: Why does resizing a passport photo change its size but never its shape, and what is the mathematics that makes this possible?

Your answer must:

- refer to the $35\text{ mm} \times 45\text{ mm}$ and $105\text{ mm} \times 135\text{ mm}$ prints and the distorted print;
- explain the role of the LSF, corresponding angles and the ratios of corresponding sides;
- explain what happens to area and to volume;
- explain what the centre of enlargement tells us about how the photo was resized.

Write your answer here:

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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Calculating and interpreting the Linear Scale Factor	Correctly calculates LSF = 3 from both dimensions, writes the formula, identifies enlargement because $LSF > 1$, and uses it accurately to find unknown lengths, with units and clear working.	Calculates LSF = 3 and uses it to find a missing length, but the formula, the enlargement reasoning or the units are incomplete.	Calculates only one ratio or makes arithmetic errors, or cannot link the number to the term Linear Scale Factor or use it to find lengths.
Explaining similarity using the two conditions	States both conditions (equal corresponding angles and equal ratios of corresponding sides) and uses the ratios 3 and 1 of the distorted print to justify clearly why it is not similar, including why similarity is multiplicative.	States both conditions and identifies that the distorted print is not similar, but the use of ratio values or the multiplicative idea is thin.	States only one condition or says the distorted print is not similar without evidence from ratios.
Relating LSF to ASF and VSF	Calculates both areas, finds the ratio 9, proves $ASF = LSF^2$ algebraically, states $VSF = LSF^3 = 27$, and clearly corrects the 'three times the paper' misconception.	Gets $ASF = 9$ and $VSF = 27$ correctly, but the algebraic proof or the misconception explanation is incomplete.	Uses LSF in place of ASF or VSF, or cannot show why the area changes by the square of the LSF.
Choosing and applying the correct scale factor in Kenyan contexts	Names the scale factor given and needed for each problem, then applies LSF, ASF or VSF correctly with accurate calculations and unit conversions.	Chooses the correct scale factor for most problems and calculates correctly, with a small slip in a unit or a missing diagnosis statement.	Uses the wrong scale factor in several problems, for example multiplying an area by the LSF.

Final Explanation of the driving question	Gives a coherent, connected explanation that uses the passport photo data, LSF, both similarity conditions, ASF, VSF and the centre of enlargement, with accurate mathematical vocabulary.	Explains the role of LSF and the similarity conditions with the photo data, but area, volume or the centre of enlargement is missing or weakly linked.	Gives a general description of resizing without referring to the LSF, the data or the similarity conditions.
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ATTRIBUTION AND LICENSING

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